

Instructor: Dr. Jim D. Pham
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Office Hrs: By Appointment
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Section: CS-116-04 (Online)

Course Description:

This intermediate-level programming course is intended for those students who already have completed an introductory programming course. It presents a comprehensive study of the C++ programming language and its role in the realm of object-oriented programming. The C++ language supports input/output streams, class constructs, inheritance, polymorphism, function and operator overloading, function and class templates, and exception handling.

Prerequisites:

Advisory: CS 102 Introduction to Computer Programming Using C++

Textbook:

C++ Programming: Program Design Including Data Structures, 8th Edition
Author: D.S. Malik
ISBN-13: 978-1-337-11756-2

Student Learning Outcomes

After completing the course, students will be able to:

1. Compare and contrast object-oriented programming with procedural programming
2. Practice creating modules using encapsulation, information hiding, inheritance, and polymorphism
3. Recognize the concept of types as a set of values together with a set of operations
4. Design and construct Exception handling methods
5. Create Linked Structures using pointers
6. Formulate procedures, functions, and iterators as abstraction mechanisms
7. Construct parameterized types (i.e. class templates in C++)
8. Propose and evaluate the separation of specification and implementation

Teaching Methods:

1. Lectures: The presentations slides and assignments will be accessed through Canvas Course Management System. See Instructional Resources below.
2. Assignments: End of chapter activities and online activities will be assigned weekly to reinforce material in the text. These assignments may require the application of various software packages.
3. Quizzes: Weekly unannounced quizzes may be given to help ensure students keep up with assigned material.
4. Participation: Student participation will be graded by the level of class participation on the Discussion Board. Each student will be required to do 2 posts per chapter on the Discussion Board.

Instructional Resources:

Canvas: Class documents, syllabus, lectures, assignments, and discussions will be posted on Canvas. You access Canvas at this URL: <https://ohlone.instructure.com>.

CANVAS CONFERENCES:

Depending on the complexity of the topics, I may conduct live lectures and labs using Canvas Conferences. Details Web Conferences will be posted on Canvas.

Email:

If you send me email, then please put **course number – name – topic** into the email **Subject** field. For example: CS 116 – Name – Issue

Discussion Board:

Your participation on the Discussion Board (i.e. Canvas) will be an important aspect of the class. Students will be required to post and respond to at least two posts on the discussion board.

Labs and Assignments:

Lab programming assignment will be assigned each week. It is the student responsibility to submit the class assignments at due date.

Exams and Tests:

There will be a) Lab Assignments; b) Online Participation; c) Quizzes; d) Midterm; e) Final Exam. You will have three attempts for each Chapter Exam and single attempts for the Midterm and Final(s). The final exam will not be comprehensive in nature. However, the instructor reserves the right to retest on material that was not appropriately comprehended. The questions will be programming, multiple-choice, true/false, and short answer fill-ins. All exams will be timed however.

Policy on Student Work:

Labs and assignments are to be done individually, unless a group project is specified. You are encouraged to talk to fellow students about concepts and methods, but you may not copy someone else's work. I adhere to the **Ohlone College Policy on Academic Integrity**.

GRADING:

Canvas Participation:	10%
Quizzes:	10%
Labs:	30%
Midterm:	20%
Final Exam:	30%

GRADING SCALE:

A = 90-100
B = 80-89
C = 65-79
D = 55-64
F = <55

Course Schedule: The following is a recommended schedule. The instructor reserves the right to make any schedule changes deemed necessary.

Session	Date	Activity
1	01/22/18	Review Basic Element of C++
2	01/29/18	Chapter 8 Array and Strings - Chapter 8 Lab Activities - Due date: 02/05/18
3	02/05/18	Chapter 9 Records - Chapter 9 Lab Activities - Due date: 02/12/18
4	02/12/18	Chapters 10 Classes and Data Abstraction - Chapter 10 Lab Activities - Due date: 02/19/18
	02/19/18	Holiday: President's Day Friday, February 16 – Monday, February 19 (weekday and weekend classes do not meet; Ohlone College Closed)
5	02/26/18	Chapter 11 Inheritance and Composition - Chapter 11 Lab Activities - Due date: 03/05/18
6	03/05/18	Chapter 12 Pointers, Classes, Virtual Functions, Abstract Classes and List - Chapter 12 Lab Activities - Due date: 03/12/18
7	03/12/18	Chapter 13 Overloading and Templates - Chapter 13 Lab Activities - Due date: 03/19/18
8	03/19/18	Midterm (Chapter 8 – Chapter 13) Chapter 14 Exception Handling - Chapter 14 Lab Activities - Due date: 03/26/18
	03/26/18	Spring Break Monday, March 19 – Sunday, March 25 (weekday and weekend classes do not meet)
9	04/02/18	Chapter 15 Recursive - Chapter 15 Lab Activities - Due date: 04/09/18
10	04/09/18	Chapter 16 Linked Lists - Chapter 16 Lab Activities - Due date: 04/16/18
11	04/16/18	Chapter 17 Stacks and Queues - Chapter 17 Lab Activities - Due date: 04/23/18
12	04/23/18	Chapter 18 Searching and Sorting Algorithms (To be determined) - Chapter 18 Lab Activities - Due date: 04/30/18

13	04/30/18	Chapter 21 Standard Template Library (STL) - Chapter 21 Lab Activities - Due date: 05/14/18
14	05/07/18	Chapter 21 Continuing and Final Exam review
15	05/14/18	Final Exam